



### \$300 million to Ebola response

The World Bank in Washington will contribute \$300 million to fight ongoing Ebola outbreak in Congo. <https://dgit.in/aug19-43>



### Manage Parkinson's symptoms

A study shows that gentle stimulation of the ear canal can help reduce symptoms of Parkinson's disease. <https://dgit.in/aug19-44>



The oobleck slime is hard when hit with a hammer (left) but can be poured as well (right).



parts cornstarch in a bowl. If you are using cornflour, you might need a bit more, and one part water. For example, if you are taking one cup of water, then use two cups of cornstarch, a bit more if cornflour. Mix food colouring in this mixture (optional). Now, take a spoon and mix the ingredients together. Note, that this will be quite hard to do and you must make sure there are no lumps. If the mixture is too watery, add more cornstarch/cornflour. That's about it! You have made oobleck slime. If you punch this slime or hit it with a hammer, it will feel hard. However, if you let the very same hammer rest on its surface for a second or two, it will sink in and the fluid can even be poured out like a liquid. We will get to the science behind this in another section.

and mix the magnet or  $\text{Fe}_2\text{O}_3$  powder in it. Mix them together until you get a consistent grey-ish colour all around. You can add some food colour if you wish to. We haven't used any in this DIY.



Next, add half a teaspoon of baking soda and mix the solution well.

After you're done mixing the magnetic solution, add half a tablespoon of lens solution and mix well. Once the solution thickens, work the slime in your hands but leave this one at a more watery consistency than the fluffy slime.

Next, bring a rare earth or neodymium magnet close to the slime. The slime should get pulled towards the magnet. You can also place the magnet in the slime, and it will eventually engulf the entire magnet. Makes for some great timelapses!

in simple PVA glue (Fevicol). Normally, borax solution is used to activate the slime, however, it is a toxic substance if ingested and can cause skin, eye or respiratory irritation, so we have used a safer alternative. Borate ion can be made when bicarbonate of soda (baking soda) and contact lens solution are mixed together. The PVA glue used in slime consists of long chains of polyvinyl acetate molecules. These chains usually slide past one another easily which keeps glue in a thick, but liquid, consistency. The mixture of PVA glue and the activator (baking soda+lens solution) causes changes in the position of the molecules in the glue. This process is known as cross-linking. The solution thickens and turns into a viscous, gooey substance,



which is known as slime, in common parlance. Slime is a Non-Newtonian fluid, meaning that they are neither a liquid nor a solid. Basically, you can pick up slime as if it were a solid, but it will flow through your fingers like a liquid. It also does not have its own shape and fits into whatever container it is put in.

Speaking of Non-Newtonian fluids, Oobleck slime is another example of one, except it is more... extreme. The harder you stir it, the thicker it gets and the more it actually resists your stirring. Also, if you hit oobleck hard enough, say with a hammer, it immediately hardens (sometimes, enough to shatter!). But the slime will still run between your fingers. There are hilarious videos online of people actually running quickly on an oobleck pool. The phenomenon behind the thickening of this substance is known as 'shear thickening'. It's a process that occurs in materials which are made up of microscopic solid particles that have been suspended in the fluid. 

## MAGNETIC SLIME



**Ingredients** - Fevicol (PVA glue), magnet powder or  $\text{Fe}_2\text{O}_3$  powder, Baking soda, Contact Lens solution, rare earth magnets or neodymium magnets, food colouring (optional)

### Procedure

Take some Fevicol in a bowl (perhaps, half of what we used in the fluffy slime)

## THE SCIENCE BEHIND SLIME

Slime is really all about polymers. A polymer is made up of a huge chain of molecules. Slime is made after a chemical reaction occurs between two primary ingredients - polyvinyl alcohol and borate ion. Both of these ingredients can easily be found in items lying around in any household. Polyvinyl alcohol is present